M534H HOMEWORK– Spring 2026

Prof. Andrea R. Nahmod

• Set 1. Due date: Thursday 02/12/2026 by 5pm

Section 1.1: 2, 3, 4.

Note Note that an equivalent manner (that what we discussed in Lecture 1) of viewing linearity is explained in Strauss' book page 2.

Assigned Background Reading: Review carefully Appendices A1. A2 and A3 in Strauss' book. We will need and use this material through the course. You will also need it to understand the first three Handouts/Section 1.3 of Strauss.

Problem: Do but do not turn in: Prove the *Second Vanishing Theorem* in A.1 page 416

• Set 2. Due date: Tuesday 02/17/2026 by 5pm

Section 1.1: 11, 12.

Section 1.2: 1, 2, 3, 4, 5, 6.

Additional Problem 1: Solve the transport equation $5u_x - 6u_y = 0$ together with the auxiliary condition that $u(x, 0) = 4x^3$.

Additional Problem 2: Solve the inhomogeneous transport equation $2u_x + 3u_y = 1$.

Assigned Reading for next week: Read carefully Handouts 1, 2 and 3 posted in the webpage. These correspond to Examples 2, 3 and 4 in Strauss' Section 1.3

• Set 3. Due date: Thursday 02/26/2026 by 5pm

Section 1.2: 7

Additional Problem 1: Solve the linear homogeneous equation $u_x - u_y = 10$

Additional Problem 2: Solve the linear homogeneous equation $u_x - u_y + u = 0$.

Specify what method you are using and explain step by step your work. Show all your work.

Additional Problem 3: a) Find the general solution to

$$2u_x + 5u_y + 29u = 0$$

Specify what method you are using and explain step by step your work. Show all your work.

b) Find the solution $u(x, y)$ to part a) that also satisfies $u(x, 0) = e^{-3x}$.

Additional Problem 4: a) Check that

$$u(x, y) = \frac{1}{4}(e^{x+2y} - e^{x-2y})$$

solves the inhomogeneous equation

$$u_x + u_y + u = e^{x+2y}.$$

b) Next use first the same method used for Additional Problem 2 (but suitably change the coefficients) to then write the general form of the solutions to

$$u_x + u_y + u = e^{x+2y}.$$

c) Find the solution to $u_x + u_y + u = e^{x+2y}$ that also satisfies $u(x, 0) = 1$.

- **Set 4. Due date: Thursday 03/05/2026 by 5pm**

Section 1.3: 6, 8.

Section 1.4: 1

Section 1.5: 1, 5, and:

Problem 6 (modified): Solve the equation $u_x + 2x\,y^2\,u_y = 0$ and find a solution that satisfies the auxiliary condition $u(0, y) = y$.

- **Set 5. Due date: Thursday 03/12/2026 by 5pm**

Reminder: Show all your work justifying your answers.

Section 1.6: 1, 2, 4, 6.

Additional Problem 1: Find the regions in $\mathbb{R}^2$ where $x^2\,u_{xx} + 4u_{xy} + y^2\,u_{yy} = 0$ is respectively elliptic, parabolic, hyperbolic. Plot these regions.

Section 2.1: 1, 2, 7.

Recall: A function $f : \mathbb{R} \to \mathbb{R}$ is said to be even if $f(x) = f(-x)$; while $f : \mathbb{R} \to \mathbb{R}$ is said to be odd if $f(x) = -f(-x)$.

Additional Problem 2 Find the solution to the wave equation $u_{tt} - 4u_{xx} = 0$, $x \in \mathbb{R}$, $t \geq 0$ with initial conditions $u(x,0) = \sin x$, $u_t(x,0) = 10$. using D'Alambert's formula. Then calculate then $u_t(0,t)$.

Additional Problem 3 Show that if $u(x,t)$ is a solution to $u_{tt} - c^2 u_{xx} = 0$, $x, t \in \mathbb{R}$ then $u(x,-t)$ is also a solution to $u_{tt} - c^2 u_{xx} = 0$.

Note. This shows that solutions to the wave equation are *time reversible*. So once you find the solution for $t > 0$ you know what the solution should be for $t < 0$ (assuming the initial conditions are given at $t = 0$.)

- **Set 6. Due date: Tuesday 03/24/2026 by 8pm (note time).**

Section 2.1: 9, 10, 11

Hint for 11: $-\frac{1}{16} sin(x+t)$ is a particular solution. Check it!.

Additional Problem 1: Find the general solution to $u_{xx} + u_{xt} - 10u_{tt} = 0$ (check whether is hyperbolic first).

Additional Problem 2: Find the solution the IVP

$$\begin{cases} u_{xx} - 6u_{xt} + 5u_{tt} = 0, & x \in \mathbb{R},\ t > 0 \\ u(x,0) = x^2 \\ u_t(x,0) = 0 \end{cases}$$

Check first whether the second order PDE is hyperbolic.

Section 2.2: 1, 2, 3

Additional Problem 3: Let $u = u(\mathbf{x}, t)$ be a solution to the wave equation $u_{tt} - \Delta u = 0$ in $\mathbb{R}^2$. Assuming that $|\nabla u| \to 0$ fast enough as $|\mathbf{x}| \to \infty$ prove that

$$E(t) = \int_{\mathbb{R}^2} |u_t|^2 + |\nabla u|^2 \, dxdy$$

is constant in t for all time t.

Additional Problem 4 Consider the wave equation in 1D with *damping*

$$u_{tt} = c^2 u_{xx} - ku - ru_t \qquad k, r > 0$$

show that the *energy functional*

$$E(t) = \frac{1}{2}\int_{-\infty}^{\infty} |u_t|^2 + c^2|u_x|^2 + k|u|^2\,dx$$

satisfies $dE/dt \leq 0$; that is *energy decreases*. Assume u and its derivatives vanish as $x \to \pm\infty$.

• Set 7. Due date: Thursday 04/16/2026 by 5pm

Section 2.3: 2, 4, 5, 6.

Hint for 4b): Do not solve explicitly. Rather prove that $u(1-x,t)$ also solves the equation and then apply the uniqueness theorem.

Section 2.4: 2, 3, 4, 8, 9, 11a) (note the statement in 11b) but don't do), 13, 15, 16.

Additional Problem 1 a) Show that the function $u(x,t) = e^{-kt}\sin(x)$ solves the heat equation $u_t - ku_{xx} = 0$.

b) Find a relationship between the constants a and b so that $u(x,t) = e^{-at}\cos(bx)$ is a solution to $u_t - ku_{xx} = 0$ (assume $\cos(bx) \neq 0$).

Bonus Problem (optional): This problem pertains the heat equation on the whole real line. Recall the Gaussian *error function*

$$\mathcal{E}\mathrm{rf}(x) := \frac{2}{\sqrt{\pi}}\int_0^x e^{-z^2}dz.$$

Prove that all solutions to the heat equation $u_t - ku_{xx} = 0$ of the form $u(x,t) = v(\frac{x}{\sqrt{t}})$, for $x \in \mathbb{R}$ and $t > 0$ have the form

$$(\dagger) \qquad u(x,t) = C_1 + C_2\,\mathcal{E}\mathrm{rf}(x)(\frac{x}{\sqrt{4kt}}).$$

Hint. First change variables $y := \frac{x}{\sqrt{t}}$, whence

$$\frac{\partial y}{\partial t} = -\frac{x}{2t\sqrt{t}}, \qquad \frac{\partial y}{\partial x} = \frac{1}{\sqrt{t}}, \qquad \text{and} \quad \frac{\partial^2 y}{\partial x^2} = 0.$$

Then prove that if $u(x,t)$ is a solution as above then $v(y)$ solves the ODE:

$$\frac{y}{2k}v'(y) + v''(y) = 0, \text{ which can be viewed as a \textit{first order} ODE for } \quad w = v'(y).$$

Solve then first $\frac{y}{2k}w(y) + w'(y) = 0$ and then by further integration find $v(y)$ expressed in terms of the Gaussian *error function.* Recall finally that $u(x,t) = v(\frac{x}{\sqrt{t}})$.

• **Set 8. Due date: Thursday 04/23/2026 by 8pm (note time).**

The following problems rely on your assigned reading of Handouts 9 and 10 in the course webpage. These correspond to parts of Strauss' Sections 3.2 (wave on the half line) and Section 3.4 (wave on the full line with a source/forcing term), so read these in parallel.

Section 3.2: 5

Section 3.4: 2

Additional Problem 1 (wave with a source) Find the solution to the following inhomogeneous wave equation on $\mathbb{R}$. Evaluate all the integrals to obtain a nice formula for the solution

$$u_{tt} - 9u_{xx} = x\,t \qquad u(x,0) = \sin(x) \qquad u_t(x,0) = 1 + x$$

Section 4.1: 2, 3.

Hints For 2) recall this is about the heat equation $u_t - ku_{xx} = 0$, $0 < x < \ell$ with zero Dirichlet boundary condition $u(0,t) = 0 = u(\ell,t)$, and initial condition $u(x,0) = 1$.

For 3) proceed as in the heat equation but keep the complex i next to $T(t)$. Recall that the solution to the ODE $T'(t) = i\lambda T(t)$ is $T(t) = Ae^{-i\lambda t}$.

The following problems are on Section 4.1. Separate variables to solve the following:.

Additional Problem 2 $u_{tt} - u_{xx} = 0$ in $0 < x < 3$ with boundary conditions $u(0,t) = u(3,t) = 0$

Additional Problem 3 $u_t = u_{xx}$ in $0 < x < \pi$ with boundary conditions $u(0,t) = u(\pi,t) = 0$

Additional Problem 4 Let g be a smooth function on $[0,1]$, $g(0) = g(1) = 0$. Consider the Dirichlet BIVP:

$$u_t - u_{xx} + 3u = 0 \text{ in } 0 < x < 1 \text{ with } u(0,t) = u(1,t) = 0 \text{ and } u(x,0) = g(x) \qquad (\oplus)$$

a) Consider the change of variables $u(x,t) = e^{-3t}v(x,t)$ and prove that v solves

$$v_t - v_{xx} = 0 \text{ in } 0 < x < 1 \text{ with } v(0,t) = v(1,t) = 0 \text{ and } v(x,0) = g(x). \qquad (\dagger)$$

b) Use the method of separation of variables to find v, the solution to (†).

c) Use a) and b) to find u, the solution to ($\oplus$).

• **Set 9. Due date: Thursday 05/08/2026 by 8pm (note time).**

Section 4.2:

Additional Problem 1 Separate variables to solve the following problem for the heat equation with zero Neumann boundary conditions: $u_t - 5u_{xx} = 0$ in $0 < x < 2$ with boundary conditions $u_x(0,t) = u_x(2,t) = 0$

Additional Problem 2 Separate variables to solve the following problem for the wave equation with zero Neumann boundary conditions: $u_{tt} - 9u_{xx} = 0$ in $0 < x < 5$ with boundary conditions $u_x(0,t) = u_x(5,t) = 0$

Section 5.1 2, 9

Hint for 9: Use the trig. identities $\cos(2x) = \cos^2(x) - \sin^2(x) = 2\cos^2(x) - 1$ to re-express the initial velocity and immediately obtain its cosine "expansion".

Section 5.2 3 (Recall the definitions of even and odd functions in page 114, formulas (3) and (4)).

Section 5.3 2a)b), $3^\dagger$ (modified below!).

Problem $3^\dagger$ Do this problem for zero Neumann boundary conditions instead of the mixed ones. That is consider instead the given wave equations with $u_x(0,t) = 0 = u_x(\ell,t)$ and the same initial data $u(x,0) = x$ and $u_t(x,0) = 0$.

Bonus Problem Let $\phi : \mathbb{R} \to \mathbb{R}$ be a periodic function with period p; that is $\phi(x+p) = \phi(x),\ \forall x \in \mathbb{R}$. Assume that ϕ is integrable on any finite interval.

(a) Prove that for any $a, b \in \mathbb{R}$

$$\int_a^b \phi(x)dx = \int_{a+p}^{b+p} \phi(x)dx = \int_{a-p}^{b-p} \phi(x)dx$$

(b) Prove that for any $a \in \mathbb{R}$

$$\int_{-p/2}^{p/2} \phi(x+a)dx = \int_{-p/2+a}^{p/2+a} \phi(x)dx = \int_{-p/2}^{p/2} \phi(x)dx$$

Note then that for any $a \in \mathbb{R}$, $\int_{-p/2}^{p/2} \phi(x+a)dx = \int_{-p/2}^{p/2} \phi(x)dx$ and thus in particular that $\int_a^{a+p} \phi(x)\,dx$ does not depend on a, as we discussed in class (section 5.2, Strauss).

Special Assignment: do but do not turn in yet

The full solution must be typed using Latex.

This problem is one of the ones that will be part of your take home final.. There will be other problems which will be assigned at the end of classes. So start working on this one now. See Course Webpage for more details on the due date of take home final exam.

Problem 1.

Consider the *initial value problem for the wave equation* on $\mathbb{R}$:

$$\begin{cases} u_{tt} - u_{xx} &= 0 \\ u(x,0) &= \phi(x) \\ u_t(x,0) &= \psi(x) \end{cases}$$

where $\phi, \psi : \mathbb{R} \to \mathbb{R}$ are two smooth given functions (data). Let $x_0 \in \mathbb{R}$ $t_0 > 0$ be fixed and suppose that $\phi(x)$ and $\psi(x)$ vanish for all x in the interval $[x_0 - t_0, x_0 + t_0]$.

Finite Propagation Speed Theorem:. The solution $u(x,t)$ to the initial value problem above vanishes for all (x,t) within $\mathcal{C}$, the domain of dependence of (x_0, t_0). Recall

$$\mathcal{C} := \{(x,t) : 0 \leq t \leq t_0 \text{ and } x_0 - (t_0 - t) \leq x \leq x_0 + (t_0 - t)\}.$$

Remark: The Theorem above is also valid in higher dimensions but for simplicity above we consider it only in one (space) dimension. In one dimension, one can trivially prove the above theorem directly using the representation formulas for the solution $u(x,t)$ in terms of the initial data which are available in one dimension. Or, one could prove it without using this explicit representation of u, but by using the *energy method* instead –as we have seen in class-. This proof is a bit harder but the advantage of the method is that it also works in higher dimensions. This Problem is then to **prove the Finite Propagation Speed Theorem using the energy method. You must follow the steps below:**

First for each $0 \leq t \leq t_0$, let $I_t := [x_0 - (t_0 - t), x_0 + (t_0 - t)]$. Note I_t is contained in the interval $(x_0 - t_0, x_0 + t_0)$.

Next define the modified energy:

$$\widetilde{E}(t) = \frac{1}{2} \int_{I_t} |u_t|^2 + |u_x|^2 \, dx$$

Note that $\widetilde{E}(t) \geq 0$ for any t and that $\mathcal{C} = \bigcup_{0 \leq t \leq t_0} I_t$.

The goal is to show that for each $0 \leq t \leq t_0$, $u(x,t) = 0$ for all $x \in I_t$. Do so by proving the following:

(1) Prove that $\widetilde{E}(t)$ is a decreasing function of t by showing that $\dfrac{d\widetilde{E}}{dt} \leq 0$

To compute the derivative in time use: (see A.3 Theorem 3 in Strauss's book p.421).

$$\frac{d}{dt}\int_{a(t)}^{b(t)} F(x,t)\,dx = \int_{a(t)}^{b(t)} \frac{d}{dt}F(x,t)\,dx + [F(b(t),t)b'(t) - F(a(t),t)a'(t)]$$

(2) Show that $\widetilde{E}(0) = 0$

(3) By (1) you then have that $\widetilde{E}(t) \leq \widetilde{E}(0)$ for any $0 \leq t \leq t_0$ and by (1) you can conclude that $\widetilde{E}(t) = 0$ for any $0 \leq t \leq t_0$. Prove then that this implies that $u(x,t) = 0$ for any $x \in I_t$ and any $0 \leq t \leq t_0$.